

Markscheme

Mathematics: analysis and approaches
Practice question bank

130. (a)

Consider $z = \frac{3 + ki}{1 - 2i}$, where k is a real number.

Find the value of k for which z is purely imaginary.

$$\text{multiply by the conjugate: } z = \frac{(3 + ki)(1 + 2i)}{5} = \frac{(3 - 2k) + (6 + k)i}{5} \quad (M1) \text{ A1}$$

$$\text{purely imaginary means the real part is zero: } 3 - 2k = 0 \quad (M1)$$

$$k = 1.5 \quad \text{A1}$$

[5 marks]